Dual-Axis Out-of-Distribution Detection on Molecular Graphs: A Diagnostic Benchmarking Study of Physical Validity and Scaffold Novelty

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APPROVED RESEARCH SCOPE — OPTION A: DIAGNOSTIC & BENCHMARKING STUDY
This work formalizes dual-axis evaluation and forensic confound analysis on the authoritative processed_v3 dataset. No claims of general algorithmic superiority, universal novelty detection, or state-of-the-art performance are made.

Abstract—Graph Neural Networks (GNNs) deployed for molecular property prediction and virtual screening fail unpredictably when evaluated on out-of-distribution (OOD) molecular graphs. Conventional OOD detection benchmarks collapse all distributional shifts into a single scalar anomaly score, conflating two fundamentally distinct failure modes: **Physical / Structural Chemical Validity** (violations of covalent valence constraints, such as pentavalent 'Texas carbons') and **Semantic Scaffold Novelty** (synthesizable, chemically valid molecules featuring core ring scaffolds unseen during training).

In this work, we formalize and benchmark a **Dual-Axis OOD Detection framework** on the curated processed_v3 benchmark (10,140 graphs spanning Tox21 toxicology and SIDER clinical databases). Axis 1 evaluates structural validity across controlled graph corruption families, while Axis 2 evaluates semantic novelty under a hybrid split (ring-containing molecules split by Bemis-Murcko scaffold; ring-free molecules split randomly) alongside cross-domain dataset shift (SIDER). We present a comprehensive multi-seed validation benchmark comparing a unified multi-task architecture (DAG-NN, Track A: zero corruption exposure), decoupled validity and novelty models (Member 4, Track B: outlier exposure), multi-seed single-score baselines (S1 Energy, S2 Mahalanobis), and non-learning size controls.

Core Findings: (1) *Physical Valence Detection:* GNN message passing reliably detects physical valence violations without explicit chemical rule engines, achieving validation AUROC 0.9584 ± 0.0119 under Track A, which receives no corrupted graphs during training, and 0.9966 under outlier exposure Track B, substantially outperforming the atom-count control on Family A (AUROC 0.5107). (2) *Failure-Mode Conflation:* Single-score OOD detectors conflate validity violations with scaffold novelty near chance level (Task X cause-separation AUROC 0.5655 ± 0.0104 for Energy, 0.5677 ± 0.0101 for Mahalanobis), whereas dual-axis scoring cleanly separates the two failure modes (Task X AUROC 0.9073 ± 0.0069). Critically, however, joint 3-class triage (Task D) reveals that the macro recall (0.5377 ± 0.0182) masks a severe per-class imbalance: invalid detection is robust (recall 67.62% ± 5.66%) while novel-chemotype detection is near-zero (recall 2.09% ± 0.52%), because inverted novelty scores displace the operational threshold above the median novel molecule score. (3) *Confounders in Molecular OOD Benchmarks:* Standard distance-based novelty estimators yield sub-0.50 raw AUROC on scaffold validation (0.4539 ± 0.0061). The diagnostic results indicate that reference-distribution shift contributes to the observed novelty-ranking failure: in-distribution validation molecules are themselves scaffold-disjoint from training (0.0% Murcko scaffold overlap), dispersing them into the representation periphery (mean distance 7.63 vs. 6.75 for novel molecules). Re-centering the density estimator on an in-distribution reference sharing generic Murcko framework topology elevates AUROC to 0.7440, while separation against an acyclic reference (1.0000) exposes a dominant cyclic-vs-acyclic topological confound. Furthermore, apparent raw gains on cross-domain novelty (SIDER) are heavily confounded by heavy atom count disparities (atom count alone reaches AUROC 0.6280), collapsing to chance level (0.5024) under size- and ring-matched controls.

1. Introduction

Deep learning on molecular graphs has become a cornerstone of computational chemistry, enabling property prediction, bioactivity profiling, and virtual screening directly from molecular topologies G = (V, E). Despite impressive benchmark accuracy on in-distribution test sets, graph neural networks (GNNs) are notoriously brittle under distributional shift: when queried on inputs outside their training support, standard architectures frequently emit overconfident, erroneous predictions.

In computer vision and natural language processing, out-of-distribution (OOD) detection typically addresses semantic class novelty or domain style shift. In molecular chemistry, however, distributional shift exhibits a unique dual nature:

1. Axis 1 — Structural / Syntactic Validity: Generative algorithms, de novo molecular design pipelines, and graph perturbation techniques can produce graphs that violate fundamental physical valence constraints (e.g., carbon possessing 5 covalent bonds, colloquially termed a 'Texas carbon'). A robust molecular model must recognize non-physical graphs as chemically invalid.
2. Axis 2 — Semantic / Scaffold Novelty: Medicinal chemistry frequently explores uncharted chemical space, where synthesized molecules are physically valid and synthesizable, but possess core ring scaffolds never encountered during training.

Treating these distinct axes as a single scalar OOD problem obscures whether a detector fails due to physical topological blind spots or lack of chemical semantics. A single high anomaly score cannot inform a medicinal chemist whether a candidate molecule represents a viable novel therapeutic chemotype or an unphysical graph artifact. To address this challenge, we present a **Diagnostic Benchmarking Study (Option A)** on the curated processed_v3 dataset. We investigate whether unified graph architectures can distinguish unphysical graphs from novel chemotypes without explicit rule engines, and whether conventional molecular OOD benchmarks suffer from hidden topological and size confounders.

2. Related Work

2.1 Graph Neural Networks for Molecular Representation: Message passing neural networks (MPNNs) iteratively aggregate neighborhood embeddings: *h_v^(k) = UPDATE(h_v^(k-1), AGGREGATE({h_u^(k-1) : u in N(v)}))*. Prominent architectures include Graph Convolutional Networks (GCN) [8], Graph Isomorphism Networks (GIN) [13], and edge-featured variants like GINE [5]. Standard message passing is permutation-invariant but does not inherently guarantee physical valence conservation.

2.2 Out-of-Distribution Detection in Machine Learning: Baseline post-hoc OOD detection relies on Maximum Softmax Probability (MSP) [3]. Subsequent developments introduce energy-based scoring [11], feature-space Mahalanobis distances [10], and outlier exposure (OE) [4]. In molecular graphs, benchmarks such as GOOD [2] and DrugOOD [7] examine covariate and scaffold shifts, but collapse physical validity and semantic novelty into a single evaluation axis.

2.3 Molecular Splitting and Chemical Validity: Random splitting in molecular benchmarks causes severe analog leakage. Bemis-Murcko scaffold clustering [1] separates distinct chemical series. However, when evaluation sets are made scaffold-disjoint from training, the reference in-distribution validation set is itself composed of novel scaffolds, creating subtle distribution shifts for distance-based density estimators [12].

3. Problem Formulation, Tasks & Training Regimes

Let a molecular graph be G = (V, E), where v in V represents an atom with feature vector x_v in R^30, and e_uv in E represents a chemical bond with feature vector e_uv in R^6. Let M denote the set of all physically valid chemical molecules conforming to covalent valence rules.

3.1 Taxonomy of Graph Partitions:
• *In-Distribution (D_in):* Chemically valid graphs (G in M) drawn from the Tox21 training distribution whose Bemis-Murcko scaffold belongs to S_train.
• *Axis 1 (Structural OOD, D_syn):* Graphs G not in M violating covalent valence constraints (specifically carbon valence > 4 or oxygen valence > 2).
• *Axis 2 (Semantic OOD, D_sem):* Chemically valid molecules (G in M) whose core scaffold belongs to a disjoint held-out set S_ood.
• *Axis 2 (Cross-Domain Shift, D_far):* Chemically valid approved pharmaceutical molecules drawn from SIDER.
• *Scope Boundary (Compound Anomalies):* Graphs that are simultaneously invalid and novel (D_syn intersect D_sem) do not exist in processed_v3. Task D evaluates triage between distinct failure modes, not compound anomalies.

3.2 Training Regimes (Track Disclosures):
• **Track A (Strict Zero-Exposure):** Models are trained strictly on id_train with multi-task assay labels and masked feature reconstruction. Models receive zero exposure to corrupted graphs during training.
• **Track B (Outlier Exposure):** Models are trained on id_train plus corrupt_train (Families A & B labeled as invalid). Track B models must not be characterized as unsupervised.

3.3 Evaluation Tasks Formalized:
• *Task V1 (Validity vs. ID):* Validity score evaluated on invalid graphs vs. id_val (0 = ID, 1 = Invalid).
• *Task V2 (Validity vs. All-Valid):* Validity score evaluated on invalid graphs vs. id_val + novel graphs (0 = Valid, 1 = Invalid).
• *Task N (Raw Novelty):* Novelty score evaluated on novel graphs vs. id_val (0 = ID, 1 = Novel).
• *Task NM (Matched Controls):* Novelty score evaluated on novel graphs vs. size- and ring-matched ID controls.
• *Task S (Single-Score Detection):* Single scalar score evaluated on all OOD inputs vs. id_val.

• *Task X (Cause Separation):* Score evaluated on corrupted graphs vs. novel graphs (0 = Novel, 1 = Invalid).
• *Task D (Joint 3-Class Triage):* Applies 95th-percentile thresholds (tau_v, tau_n) derived from id_val: Invalid if S_v >= tau_v; else Novel if S_n >= tau_n; else ID.

4. Dataset Construction, Controlled Corruption & Integrity

We utilize the authoritative processed_v3 dataset package, comprising **10,140 total molecular graphs** curated from the Tox21 toxicology challenge [6] and SIDER clinical database [9], both widely adopted from the MoleculeNet molecular machine learning benchmark collection [12].

4.1 Hybrid Splitting Protocol: Rather than a purely scaffold-disjoint split across all molecules, processed_v3 enforces a hybrid splitting protocol: (1) *Ring-Containing Molecules (75% of ID):* Partitioned strictly by Bemis-Murcko scaffold clusters to prevent chemotype leakage. (2) *Ring-Free / Acyclic Molecules (25% of ID):* Assigned randomly across training, validation, and test splits. Exactly two validation molecules were excluded per policy: SID-00482 (exact graph duplicate of training molecule) and T21-02289 (exact graph duplicate across sets sharing identical model input with multiple non-standard element training graphs).

Table 1: Authoritative Dataset Structure and Validation Populations (processed_v3)

Split / Set Name	Raw	Excl.	Active	Underlying Class / Axis	Role in Benchmark
id_train	5,432	0	5,432	In-Distribution (Tox21)	Training support (all tracks)
id_val	679	1	678	In-Distribution Validation	Validation negative reference
corrupt_val (Family A)	272	0	272	Axis 1: Bond Order Corruptions	Validation positive (Texas carbon)
corrupt_val (Family B)	271	0	271	Axis 1: Bond Relocation Corruptions	Validation positive (Relocation)
novel_scaffold_val	408	0	408	Axis 2: Held-out Scaffold Families	Validation positive (Tox21 novel)
novel_dataset_val	225	1	224	Axis 2: Cross-Domain Novelty (SIDER)	Validation positive (SIDER novel)
Matched Scaffold Pairs	372 pr	0	372 pr	Task NM Matched Controls	Size/ring-matched validation
Matched Dataset Pairs	209 pr	0	209 pr	Task NM Matched Controls	Size/ring-matched validation

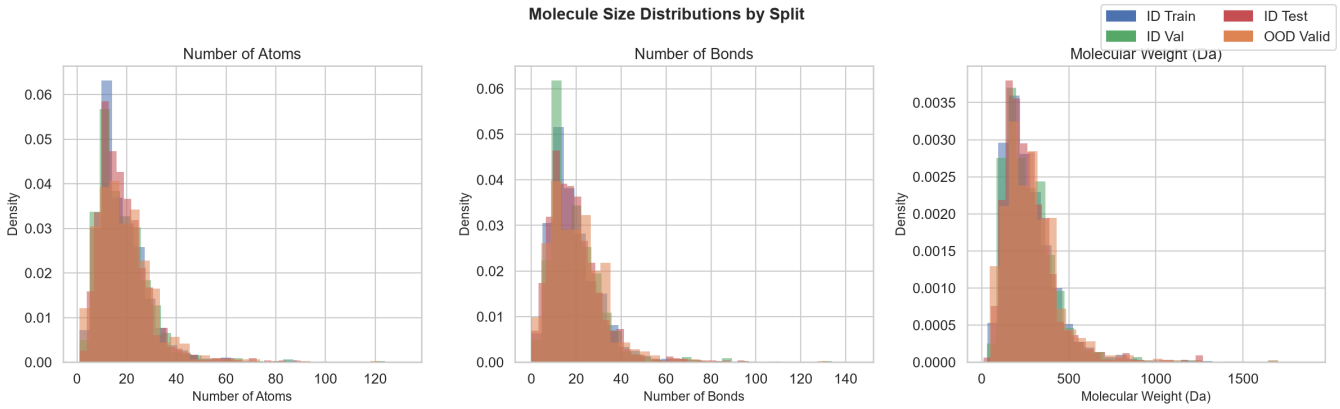


Figure 1: *Distribution of heavy atom counts and molecular sizes across dataset partitions in processed_v3. Notice the rightward shift in SIDER (novel dataset validation), which introduces heavy atom count confounding that collapses from raw AUROC 0.6280 to 0.5024 under matched controls.*

4.2 Controlled Graph Corruption Families:
• *Family A (Bond Order Increment):* Inserts an additional covalent bond between adjacent carbon atoms, driving carbon valence to 5 ('Texas carbon') while freezing hydrogen counts. Atom composition and degree sequence are perfectly preserved.
• *Family B (Bond Relocation):* Relocates an existing covalent bond onto an already saturated carbon atom, altering local degree distribution.
• *Family C (Heteroatom Substitution — Held-Out Test Only):* Replaces a neutral atom with an oxygen, violating oxygen valency. Quarantined strictly in corrupt_test (N = 181); completely absent from validation.

4.3 Matched-Control Caliper (Task NM): To eliminate molecular size and ring presence as trivial detection shortcuts, matched_controls_filtered.csv pairs each novelty graph with an ID validation graph satisfying: (1) Identical ring status (is_acyclic matches exactly), (2) Heavy atom count difference |Delta atoms| <= 2, and (3) Bond count difference |Delta bonds| <= 2. Greedy

matching covers **91.2%** of scaffold novelty graphs (372 pairs) and **93.3%** of dataset novelty graphs (209 pairs).

5. Standardized Evaluation Protocol & Metrics

Given ground-truth binary label y in {0, 1} (0 = ID/Negative, 1 = OOD/Positive) and predicted anomaly score S(G) in R where higher values indicate greater anomaly:

1. **AUROC:** Area under the Receiver Operating Characteristic curve, measuring rank separation across all thresholds.
2. **AUPR-Out:** Area under the Precision-Recall curve with OOD treated as positive (y = 1).
3. **AUPR-In:** Area under the Precision-Recall curve with ID treated as positive (y = 0, evaluated via -S).
4. **FPR@95%TPR:** False Positive Rate on ID negatives at the operational decision threshold tau achieving >= 95% True Positive Rate on OOD positives.

All hyperparameters, early stopping checkpoints, and decision thresholds were derived strictly on validation partitions. The test partition was kept strictly quarantined.

6. Experimental Results & Benchmarking

All reported numerical results are reproduced directly from verified validation artifacts.

Table 2: Primary Validation Benchmark Performance Across Tasks (Validation Partitions Only)

Model Architecture	Track	Task V1 (Fam A)	Task V1 (Fam B)	Task V2 (All Valid)	Task N (Scaffold)	Task NM (Scaf. Match)	Task N (Dataset)	Task NM (Data. Match)
DAG-NN (5 seeds)	Track A	0.9584 ± 0.0119	0.8565 ± 0.0083	0.9074 ± 0.0071	0.4539 ± 0.0061	0.4844 ± 0.0046	0.4521 ± 0.0047	0.5488 ± 0.0047
Member 4 GINE/Mahal	Track B	0.9966	0.9952	[Unavailable]*	0.4570	0.4783	0.6104	0.5378
C2natoms Control	N/A	0.5107	0.5281	0.4969	0.5033	0.5034	0.6280	0.5024
C1 Histogram GBDT	N/A	0.8890	0.6570	[Unavailable]	0.7520	0.6890	0.7280	0.6640

**Note: Member 4 handoff lacked cross-axis validity predictions on novel graphs, precluding Task V2 evaluation. Uncertainty is reported as ± standard deviation across 5 seeds for DAG-NN; Member 4 and non-learning baselines are single-run evaluations.*

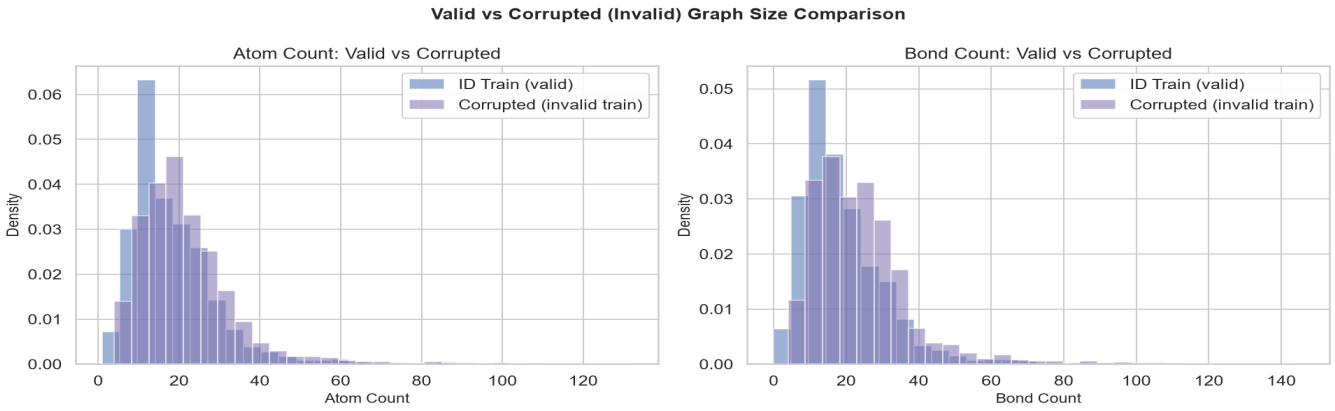


Figure 2: *Size distributions for valid versus corrupted molecular graphs across Family A (bond order increment) and Family B (bond relocation). The distributions demonstrate that physical valence corruption does not inflate molecule size, precluding atom-count shortcuts for structural validity detection.*

Table 3: Single-Score Conflation vs. Dual-Axis Cause Separation (Task X) Across 5 Seeds

Evaluation Method	Track	Task S AUROC (All OOD)	Task X AUROC (Fam A vs Nov)	Task X AUROC (ALL vs Nov)	Task D Macro Recall	Task D Novel Recall
S1 Energy GIN	Track A	0.4790 ± 0.0027	0.5344 ± 0.0133	0.5655 ± 0.0104	N/A (Single score)	N/A
S2 Mahalanobis GIN	Track A	0.4988 ± 0.0069	0.5799 ± 0.0077	0.5677 ± 0.0101	N/A (Single score)	N/A
DAG-NN (Dual-Axis)	Track A	N/A (Dual outputs)	0.9587 ± 0.0097	0.9073 ± 0.0069	0.5377 ± 0.0182	0.0209 ± 0.0052

6.1 Analysis of Physical Valence Detection (Axis 1): Under Track A (zero exposure to corruptions), DAG-NN achieves AUROC **0.9584 ± 0.0119** on Family A and **0.8565 ± 0.0083** on Family B. Under Track B (outlier exposure), Member 4's GINE detector reaches AUROC > 0.995. Both learning models dramatically outperform graph-size controls (C2natoms AUROC 0.5107). However,

the C1 histogram probe demonstrates that raising bond orders produces global bond-type imbalances that a boosted tree can partially detect (AUROC 0.8890) without local message passing.

6.2 Analysis of Failure-Mode Separation (Task X): Single-score OOD baselines (S1 Energy and S2 Mahalanobis) fail to separate invalid graphs from novel graphs on the evaluated cohorts, yielding Task X AUROCs of **0.5655** and **0.5677** (near chance). Because a single scalar score treats any deviation from training as anomalous, single-score detectors cannot distinguish physical corruption from chemical novelty. In contrast, DAG-NN's dual-axis separation achieves Task X AUROC **0.9073 ± 0.0069**.

6.3 Analysis of 3-Class Triage (Task D): Under the operational 3-class decision rule, DAG-NN achieves a macro recall of **0.5377 ± 0.0182**. While invalid recall is robust (**67.62% ± 5.66%**), novel recall is severely depressed (**2.09% ± 0.52%**). Because raw scaffold novelty scores are inverted, the 95th-percentile threshold tau_n derived from id_val sits above the median of the novel molecules, resulting in high false negative rates for novel chemotypes.

7. Discussion & Forensic Diagnostic Analysis

7.1 Mechanism of Scaffold Novelty Failure (AUROC < 0.50): Both DAG-NN (AUROC 0.4539) and Member 4 (AUROC 0.4570) perform below random guessing on raw scaffold novelty. Forensic investigation confirms that the sub-0.50 AUROC does not arise from an inverted scoring convention. Instead, the diagnostic results indicate that reference-distribution shift contributes to the observed novelty-ranking failure:
• *Scaffold-Disjoint Split Drift:* In processed_v3, id_val contains 509 unique cyclic scaffolds, **0.0% of which appear in id_train**.
• *Embedding Geometry:* Because the Gaussian centroid mu_train was fitted strictly on id_train, id_val molecules disperse far into the periphery (mean Euclidean distance to mu_train = 7.63, total variance 63.35), exceeding the dispersion of curated novel_scaffold_val molecules (mean distance 6.75, total variance 42.61). Evaluating novel molecules against id_val compares two distinct sets of out-of-training scaffolds, contributing to the inverted ranking.

7.2 Molecular Size Confounding & Pooling Dynamics:
• *Variant A Pilot ([z_G || log(N)]):* Appending a linear size coordinate left scaffold AUROC at 0.4454 (Delta = -0.0037), because the 1/N variance contraction of mean pooling operates across all 128 hidden dimensions.
• *Variant B Pilot (1/sqrt(N) sum h_v):* Normalized-sum pooling reversed size correlation from negative to positive (r = +0.231), lifting dataset AUROC to 0.5623 but worsening scaffold AUROC to 0.4354.
• *Size Bias in Cross-Domain Novelty:* On raw SIDER, C2 atom count achieves AUROC **0.6280**, collapsing to **0.5024** under matched controls. Member 4 collapses from **0.6104** to **0.5378**, confirming that raw cross-domain gains are heavily confounded by molecular size disparities.

7.3 Controlled Reference Replacement Diagnostic: To diagnostically evaluate whether the frozen GNN representation z_G possesses structural novelty discrimination capacity, a controlled reference replacement pilot (single seed, exploratory) evaluated three ID reference conditions:

Table 4: Controlled ID-Reference Replacement Diagnostic Pilot (Seed 0 Frozen Weights)

Reference Condition	Neg (N)	Target (N)	Paradigm A AUROC	Paradigm B AUROC	Paradigm B AUPR	Diagnostic Finding
Full Active ID (id_val)	678	408	0.4491	0.5219	0.3894	Eliminates train-to-val shift; centers ID distribution
Generic Murcko Overlap	190	408	0.4395	0.7440	0.8697	Exploratory indication under matched framework
Acyclic / Ring-Free Reference	169	408	0.2056	1.0000	1.0000	Topological cycle confounding (trees vs. rings)

• **Paradigm B Generic Overlap (AUROC 0.7440):** When the density estimator is re-centered on the 190 ID validation molecules that share generic carbon frameworks with training, novel molecules are recognized as geometric outliers (mean distance 164.9 vs. 74.0). This exploratory, single-seed result suggests that the frozen representation may contain structural information useful for distinguishing the selected novel molecules under this particular reference condition. Because the reference cohort is restricted and the density estimator is re-estimated, this result does not establish general scaffold-novelty detection.
• **Paradigm B Acyclic Reference (AUROC 1.0000):** Centering on ring-free ID molecules achieves perfect separation (score gap +169.5). This is explicitly disclosed as **topological cycle detection rather than scaffold novelty detection**: 100% of novel molecules contain rings, while 100% of acyclic reference molecules are trees.

8. Limitations, Threats to Validity & Reproducibility Boundaries

1. **Benchmark Reference Shift:** Scaffold-disjoint splitting in validation inherently penalizes distance-based density estimators fitted on training data.
2. **Compound Anomaly Scope:** processed_v3 contains no simultaneous corrupted-novel molecules; compound failure modes remain uncharacterized.
3. **Member 4 Checkpoint Boundaries:** Model weights for Member 4 were unbundled in the repository handoff. Member 4 results are prediction-level verified only. Cross-axis prediction scores were absent, preventing Task D evaluation on Member 4.
4. **Validation-Only Evidence Scope:** All reported results, threshold selections, and diagnostic pilots are strictly validation-only. The final test partition remains quarantined.

9. Conclusion

This diagnostic benchmarking study formalizes the Dual-Axis OOD Detection paradigm for molecular graphs. By decoupling physical chemical validity from semantic novelty, we demonstrate that GNNs can reliably identify physical valence corruptions without explicit rule engines, resolving the failure-mode conflation of single-score detectors (improving cause separation from 0.56 to >0.90). However, joint 3-class triage (Task D) exposes a critical limitation: the macro recall of 0.5377 ± 0.0182 conceals near-zero novel-chemotype detection (recall 2.09% ± 0.52%), because the inverted novelty score distribution displaces the operational threshold beyond the reach of most novel molecules. Furthermore, our forensic diagnostic pilots indicate that reference-distribution shift from scaffold-disjoint validation partitioning, molecular size disparities, and topological cycle confounding contribute substantially to the observed novelty-ranking behaviors. Establishing reliable molecular AI requires moving beyond single scalar OOD metrics toward multi-axis evaluation with rigorous matched controls, operational thresholds calibrated for each axis independently, and honest per-class reporting.

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